Mathematics

International Physics Olympiad · Lecture + Problems

Introductory study guide · 10 original problems

> Organized around the IPhO syllabus. This guide is not exhaustive; consult the syllabus study PDF and the IPhO syllabus webpage for the full scope and qualifications.

4.1 Algebra

Factor and solve equations, retaining only physically meaningful roots. Arithmetic and geometric series simplify repeated contributions; for ratio $r \neq 1$, $\Sigma_{k=0}^{n-1} r^k = (1-r^n)/(1-r)$.

4.2 Functions

Use trigonometric identities, inverse functions, exponentials and logarithms. For example, ln(ab) = ln a + ln b and sin(a+b) = sin a cos b + cos a sin b.

4.3 Geometry and stereometry

Use radians in calculus, similarity, sine and cosine rules, and standard areas and volumes. Recognize circles, ellipses, parabolas and hyperbolas as conic sections.

4.4 Vectors

Resolve vectors into components. The dot product is ab cos θ and the cross product has magnitude ab sin θ. Their signs and directions encode work and torque; the derivative of a rotating vector may change direction without changing its magnitude.

4.5 Complex numbers

Write z = a + ib = re^(iφ). Multiplication adds phases, while division subtracts them. Complex amplitudes simplify sinusoidal steady-state circuits; the observable is the real part.

4.6 Statistics

For independent observations, the mean is $\bar{x} = \Sigma x_i/n$. The sample standard deviation is $s = \sqrt{[\Sigma(x_i-\bar{x})^2/(n-1)]}$; the estimated standard error of the mean is $s/\sqrt{n}$.

4.7 Calculus

A derivative measures local rate; a definite integral accumulates change. Apply product and chain rules, use initial conditions to determine integration constants, and interpret the gradient as the direction of steepest increase.

4.8 Approximate and numerical methods

For $|x| \ll 1$, $(1+x)^{\alpha} \approx 1+\alpha x$. Bisection brackets a root by repeatedly halving an interval with opposite endpoint signs. The trapezoidal rule approximates an integral by a sum of trapezoid areas.

Problems

Try each problem before opening the worked solutions. Exercises are original MAIS PAC teaching material, not official IPhO problems.

1. Solve $x^2 - 5x + 6 = 0$.
2. Solve e^(−t/2) = 1/4 for t.
3. Find the area of a circle of radius 3 m.
4. For a = (1, 2) and b = (3, 4), find a · b.
5. Express $(1+i)^2$ in algebraic form.
6. Measurements are 2, 4 and 6. Find their mean and sample standard deviation.
7. If $x(t) = 3t^2 + 2$, find velocity and displacement from t = 0 to t = 2.
8. Approximate $\sqrt{1.04}$ to first order in 0.04.
9. Challenge 1 — A positive solution of $x + \varepsilon x^2 = 1$ is close to 1 when $|\varepsilon|$ is small. Find it to first order in ε.

10. Challenge 2 — An ideal series RC circuit has resistance R, capacitance C and angular frequency ω. Find the magnitude of the voltage across the capacitor divided by the source voltage.